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Source: Journal of Medical Entomology, 43(6):1178-1184.
Published By: Entomological Society of America
DOI: http://dx.doi.org/10.1603/0022-2585(2006)43[1178:IOTRVO]2.0.CO;2
URL: http://www.bioone.org/doi/full/10.1603/0022-2585%282006%2943%5B1178%3AIOTRVO%5D2.0.CO%3B2

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Impact of the Rift Valley on Restriction Fragment Length Polymorphism Typing of the Major African Malaria Vector *Anopheles funestus* (Diptera: Culicidae)

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J. Med. Entomol. 43(6): 1178–1184 (2006)

ABSTRACT *Anopheles funestus* Giles (Diptera: Culicidae) is one of Africa's major malaria vectors. To understand population structure within *An. funestus*, various molecular markers have recently been developed. We investigated the impact of the Rift Valley on one such molecular marker, a polymerase chain reaction-restriction fragment length polymorphism (RFLP). This system distinguishes *An. funestus* populations into types M, W, and MW. Generally, these types correlate between geographical location and RFLP type, whereby M associates essentially with eastern Africa, W with western and central Africa, and MW with southern Africa. In total, 606 specimens were screened from 12 countries. Specimens from southern African localities were of the expected MW-type with the exception of the sample from Malawi, which showed all three types. The specimens from Tanzania showed the M- and MW-types, whereas specimens from Kenya showed M- and W-types. We also recorded different RFLP types not recorded previously. Type Y was recorded in Malawi, and Z-type was recorded from four localities: Angola, Malawi, Ghana, and Zambia.

KEY WORDS *Anopheles funestus*, restriction fragment length polymorphism types, malaria, population structure

Anopheles funestus Giles (Diptera: Culicidae) has been shown to be one of the most important vectors of malaria in sub-Saharan Africa, sometimes more so than *Anopheles gambiae* Giles and *Anopheles arabiensis* Patton (Gillies and De Meillon 1968, Shiff et al. 1995, Fontenille et al. 1997, Temu et al. 1998, Coetzee and Fontenille 2004). It is one of nine members within the *An. funestus* group (Gillies and Coetzee 1987, Cohuet et al. 2003). An understanding of the genetic population structure and the nature of barriers to gene flow in *An. funestus* is important to have a better understanding of malaria epidemiology and is crucial to effective vector control, especially because emerging insecticide resistance has been recognized (Hargreaves et al. 2000). Understanding the geographical barriers to gene flow within and between genetically differentiated units is important for current vector control efforts involving insecticides, because it allows prediction of the spread of genes conferring insecticide resistance. Despite the importance as a vector, research on the *An. funestus* group, including studies on the population genetic structure, was neglected for many years. This neglect was due to various factors, including difficulties with species identification and laboratory colonization. These difficulties have recently been overcome with the development of several molecular tools to accurately identify the species (Koekemoer et al. 2002; Garros et al. 2004a,b; Cohuet et al. 2003) and the establishment of stable colonies (Hunt et al. 2005).

Little is known about the population genetic structure of *An. funestus* compared with *An. gambiae* s.s., which received much more attention. However, it has been demonstrated that a high level of cytogenetic, bionomic, and molecular heterogeneity was observed within *An. funestus* over Africa, which may demonstrate potential population subdivisions. Three different chromosomal forms, namely, Dielmo, Skouvar, and Wassadou, were described in Senegal based on polytene chromosome inversion polymorphisms (Lochouarn et al. 1998). Two additional chromosomal forms, Folonzo and Kiribina, have been identified in Burkina Faso (Costantini et al. 1999). The Folonzo form is chromosomally similar to eastern and southern African populations. This work also showed that the Folonzo form seems to be a better malaria vector than the Kiribina form. It has been hypothe-

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0022-2585/06/1178–1184$04.00/0 © 2006 Entomological Society of America

sized that they are reproductively isolated and are in the process of incipient speciation based on an observed lack of heterozygous chromosomal types while in sympatry (Costantini et al. 1999). In East Africa, Kamau et al. (2003) suggested that the Rift Valley acts as a barrier to gene flow among populations of *An. funestus* from Kenya based on chromosomal inversion polymorphisms. However, in contrast to western Africa, inversion polymorphism is lower in East and South Africa and there is no evidence of distinct chromosomal forms.

Despite the considerable geographical structure shown by chromosomal inversions that may indicate the presence of several reproductive units within what is currently recognized as *An. funestus* s.s., microsatellite, mitochondrial DNA (mtDNA) and rDNA markers show relatively low levels of population structure (Cohuet et al. 2004, Coetzee and Fontenille 2004). In Cameroun and Senegal, microsatellite studies showed no evidence of restricted gene flow within inversions that distinguish the chromosomal forms, relative to other regions of the genome, the chromosomal forms may be maintained by environmental selection alone (Cohuet et al. 2004). In eastern Africa, Braginets et al. (2003) and Temu et al. (2004), by using microsatellite DNA markers, showed, respectively, that there is significant genetic differentiation between *An. funestus* populations from coastal and western Kenya and a lack of gene flow between eastern/southern African *An. funestus* populations and those from Malawi. Recently, a restriction fragment length polymorphism (RFLP) technique was used to study *An. funestus* populations (Garros et al. 2004a). In this assay, a variable domain (D3) within the 28S ribosomal gene is amplified before digestion with HpaII/MspI. The 28S gene is located in the rDNA, which is present as a tandem repeat (>100) in the insect genome and is located on the X-chromosome (Paskewitz et al. 1993).

More recently, Michel et al. (2005a,b) demonstrated distinct differences in southern African *An. funestus* based on mitochondrial polymorphisms. Using microsatellite DNA markers, they defined three distinct clusters. The first cluster was "Africa, east of the Great Rift Valley" and included Tanzania, Malawi, Mozambique, and Madagascar. The second cluster included West Africa (Burkina Faso, Mali, Nigeria and Kenya [west of the Rift Valley]), and the third cluster included Gabon and coastal Angola. It was clearly demonstrated from this study that there is a definite division between eastern and western populations (Michel et al. 2005a,b), similarly to the clusters defined by Garros et al. (2004a).

In this study, we aimed to increase the number of typed specimens by using RFLPs from some of the localities reported in Garros et al. (2004a) (Angola, Mozambique, South Africa, and Kenya) to increase the sample size for each of these localities as well as to add supplementary additional sites in eight other countries (Côte d'Ivoire, Ghana, Malawi, Mali, Nigeria, Tanzania, Uganda, and Zambia). We also investigated the impact that the Rift Valley in East Africa has on RFLP profiles in Kenya (Ahero, Kanyaboli, Kisii, Rota, Magaoni, Jaribuni, and Taveta), Malawi (Karonga and Nkhotakota), and Tanzania (Geita and Bagamoya).

Materials and Methods

Mosquito Specimens and Sampling Sites. Mosquitoes used in this study were field specimens collected from 12 African countries (Table 1). They were preserved on desiccated silica gel before analysis. All samples were collected indoors as adult females.

DNA Extraction and A Priori Identification. Specimens used for this study were morphologically identified using the keys of Gillies and Coetzee (1987). Those identified as members of the *An. funestus* group were identified to species level by using the species-specific PCR assay of Koekemoer et al. (2002). Genomic DNA was extracted from individual adult mosquitoes according to the procedure of Collins et al. (1987) and resuspended in 200 μl of TE buffer. A negative control was included with every set of extractions (called no-DNA) as well as a positive control (colonized *An. funestus* from Mozambique, labeled FUMOZ).

Amplification of the D3 Region and Restriction Digestion. RFLP assays were done according to Garros et al. (2004a). Briefly, after polymerase chain reaction (PCR) amplification, 10 μl of PCR-amplified template was restricted using HpaII enzyme. Samples were then subjected to electrophoresis on a 3% agarose-gel in 1X TBE buffer, pH 8. Each PCR reaction was accompanied by two negative controls, the first using the "no-DNA" extraction as template and the second containing an aliquot of the PCR mixture excluding template.

Sequencing and Analysis. Samples were sequenced directly from PCR products. Each new or known type (Y, Z, and MW) was sequenced three times to ensure PCR mutations were not incorporated in the sequence. Sequencing was done through Inqaba Biotechnical Industries Pty Ltd., Pretoria, South Africa. Sequencing reactions were carried out using the BigDye version 3.1 terminator cycle sequencing kit (Applied Biosystems, Foster City, CA), and the samples were loaded into a Spectrumedix SCE2410 genetic analyzer (SpectruMedix LLC, Bellefonte, PA). GenBank accession numbers are as follow: Y-type, DQ407757; MW-type, AY259149.1 (Garros et al. 2004 a); W-type, AY259153.1 (Garros et al. 2004 a); M-type, AY259151.1 (Garros et al. 2004 a); and Z-type, DQ407756. A phylogenetic tree was generated using MegAlign software (Lasergene 6, DNASTAR, Inc., Madison, WI) and is rooted.

Mapping of Study Sites. A global positioning system record of the geographical location of all study areas was taken (Table 1). The map of the genotypic locations of *An. funestus* was produced using the ArcGis 8.1 software (ESRI, Redlands, CA).

Table 1. Specimens used in this study (in bold) and in Garros *et al.* (2004a), their geographical coordinates, sample size, and number of specimens per molecular form

Map no.	Country	Collection site	Map reference	*n*	Molecular form					Reference
					Z	M	MW	Y	W	
1	Angola	Calueque	17° 20′ S, 14° 44′ E	**35**	**1**		**34**			This study
1		Calueque	17° 20′ S, 14° 44′ E	14			14			Garros *et al.* (2004)
2	Burkina Faso	Boromo	11° 30′ N, 2° 54′ W	15					15	Garros *et al.* (2004)
3	Cameroon	Simbock	3° 50′ N, 11° 30′ E	21					21	Garros *et al.* (2004)
4	Cote d'Ivoire	Petionara	7° 42′ N, 5° 00′ W	**16**					**16**	This study
5	Ghana	Jimiso	6° 16′ N, 1° 28′ W	**24**	**1**				**23**	This study
6	Kenya	Ahero	0° 10′ S, 34° 55′ E	**52**					**52**	This study
7		Kanyaboli	0° 02′ N, 34° 12′ E	**53**					**53**	This study
8		Kisii	0° 46′ S, 34° 56′ E	**42**					**42**	This study
9		Rota	0° 08′ S, 34° 36′ E	**29**					**29**	This study
10		Magaoni	4° 11′ S, 39° 26′ E	**26**		**26**				This study
10		Magaoni-Kwale	4° 11′ S, 39° 26′ E	5		5				Garros *et al.* (2004)
11		Jaribuni	3° 38′ S, 39° 45′ E	**21**		**21**				This study
12		Taveta	3° 22′ S, 37° 39′ E	**30**		**30**				This study
13	Madagascar	Ambohimena	18° 04′ S, 46° 42′ E	56		56				Garros *et al.* (2004)
14	Malawi	Karonga	9° 09′ S, 33° 09′ E	**36**				**36**		This study
15		Nkhotakota	13° 30′ S, 34° 00′ E	**9**	**1**	**6**		**1**	**1**	This study
16	Mali	Morilla Mine	11° 30′ N, 6° 50′ W	**1**					**1**	This study
17	Mozambique	Boane	26° 02′ S, 32° 19′ E	**42**			**42**			This study
18	Nigeria	Abeokuta	7° 08′ N, 3° 25′ E	**9**					**9**	This study
19	Senegal	Kedougou	12° 33′ N, 12° 11′ W	25					25	Garros *et al.* (2004)
20	South Africa	Mpumalanga	25° 26′ S, 31° 57′ E	**38**			**38**			This study
21		Kwazulu/Natal	27° 02′ S, 32° 16′ E	**22**			**22**			This study
21		Kwazulu/Natal	27° 02′ S, 32° 16′ E	25			25			Garros *et al.* (2004)
22	Tanzania	Geita	2° 52′ S, 32° 10′ E	**48**			**48**			This study
23		Bagamoya	6° 26′ S, 38° 55′ E	**18**		**18**				This study
24	Uganda	Kakira	1° 00′ N, 32° 00′ E	**5**					**5**	This study
25		Jinja	00° 18′ N,32° 00′ E	**14**					**14**	This study
26		Tororo	00° 41′ N,34° 10′ E	**28**					**28**	This study
27	Zambia	Ndola	30°48′ S, 22°10′ E	**7**	**1**		**2**	**4**		This study
28		Chipata	13° 06′ S, 32°06′ E	**1**			**1**			This study
	Total for this study			**606**	**4**	**101**	**187**	**41**	**273**	

Numerical numbers next to collection site refer to Fig. 3.

Results

In total, 606 specimens that originated from 12 countries in Africa were positively identified as *An. funestus* and used in the RFLP assay (Table 1). All specimens from West Africa ($n = 50$; Côte d'Ivoire, Ghana, Mali, and Nigeria) gave the specific W-type, except one specimen from Ghana that gave a new RFLP type that was labeled Z-type (Table 1; Fig. 1, site 5). Three fragments characterized this type, the 100-bp fragment found in all *An. funestus* and the 195-bp fragment found in W-type. However, the 150-bp fragment found in W-type was absent and was replaced with a 140-bp fragment. Sequence analysis on Z-type (DQ407756) revealed two HpaII cutting sites: 84 and 220 bp.

In total, 145 specimens from southern Africa (Angola, Mozambique, South Africa, Zambia) were tested and all gave the diagnostic MW-type, except one specimen from Zambia that showed the Z-type. Among the other seven specimens from Zambia ($n = 8$), four specimens gave a similar MW-type profile but with distinctly different banding intensities. These specimens were preliminary labeled as Y-type (Fig. 1). Sequencing of this type (DQ407757) showed only three nucleotide substitutions with *An. funestus* from Mozambique (AY259149.1), and alignment of nucleotide sequences of the ITS2 region between the MW-, Z-, and Y-type is illustrated in Fig. 2A.

Rift Valley Populations. Specimens from Kenya, Malawi, Tanzania, and Uganda (Table 1; Fig. 3) were analyzed to study the effect of the Rift Valley on

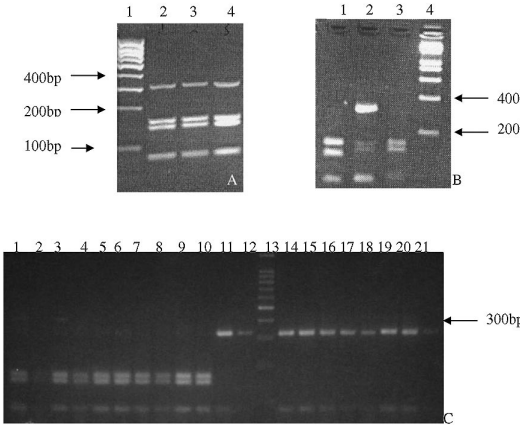


Fig. 1. RFLP patterns showing four different types. (A) Lane 1, molecular marker (100-bp ladder; Invitrogen, Carlsbad, CA); lanes 2–4, MW-type from *An. funestus* from Mozambique. (B) Lane 1, Z-type; lane 2, Y-type; lane 3, W-type; and lane 4, molecular marker (Hyperladder, Celtic Molecular Diagnostics, Cape Town, South Africa). (C) Lanes 1–10, W-type; lanes 11–12 and 14–21, M-type; and lane 13, molecular marker (100 bp; Promega, Madison, WI).

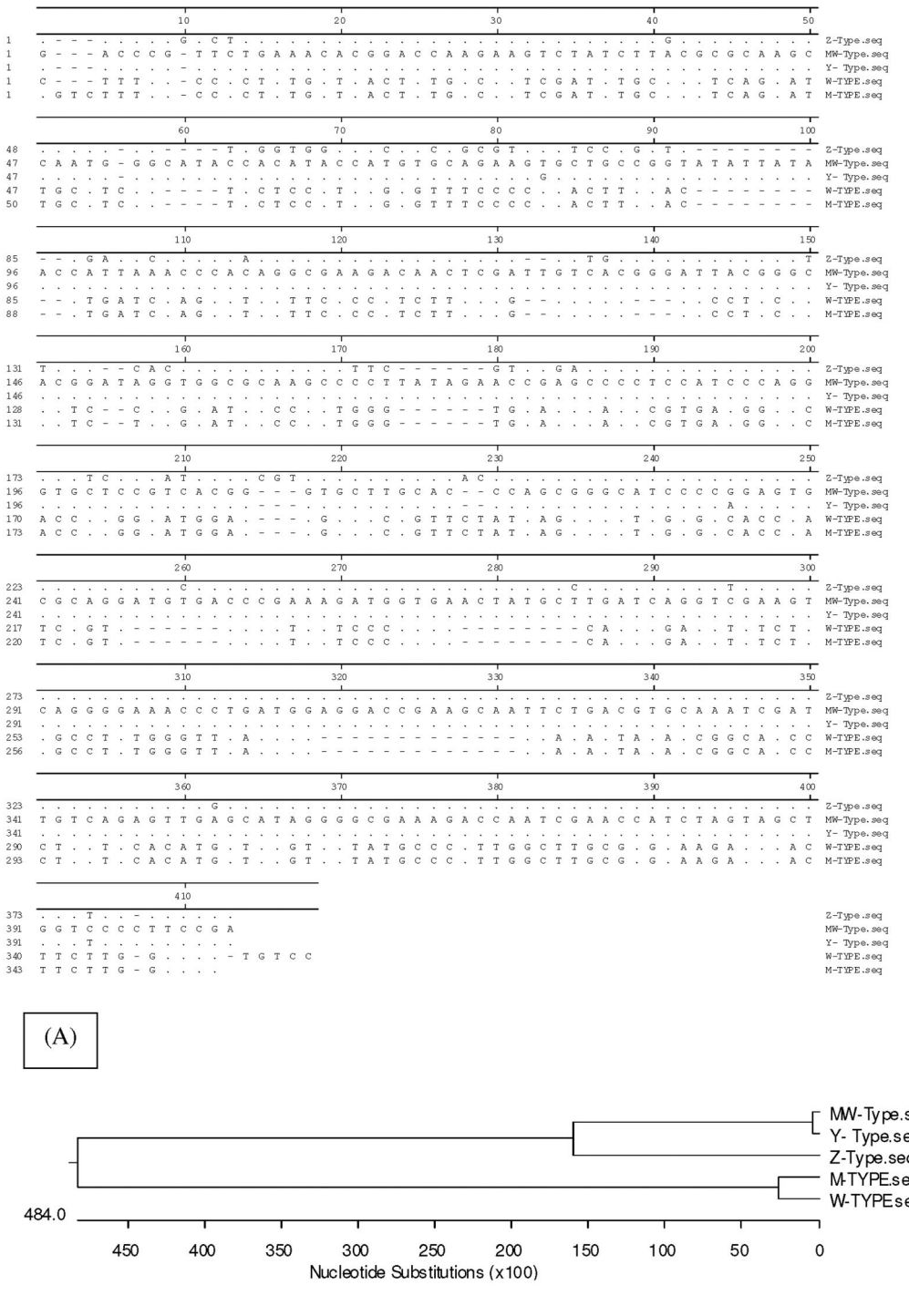


Fig. 2. (A) Alignment of the nucleotide sequence of the D3 domain in *An. funestus* for the MW-type, M-type, W-type, Z-type, and Y-type. (.) indicates same nucleotide as MW-type, and (-) indicates the absence of a nucleotide at that specific position. (B) Phylogenetic tree predicted from the multiple sequence alignment. The length of each pair of branches represents the distance between sequence pairs. Units at the bottom of the tree indicate the number of substitution events. A scale is indicated below the tree as the number of "nucleotide substitutions."

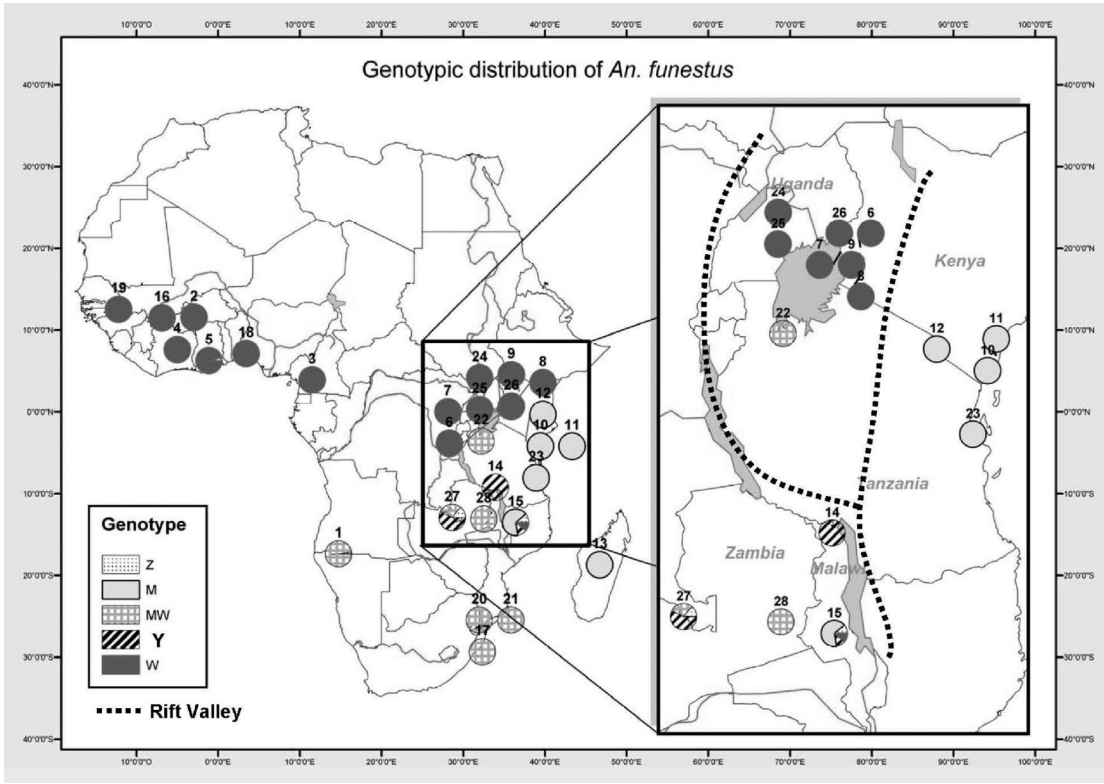


Fig. 3. Distribution of the five genotypes of *An. funestus* populations in 16 African countries. Numbers refer to study sites (see Table 1). Dotted line indicates relative position of the Rift Valley.

various *An. funestus* populations. Uganda, on the west side of the Rift Valley, showed the West African W-type ($n = 47$). Kenyan specimens ($n = 176$) from four localities (Ahero, Kanyaboli, Kisii, and Rota) located on the western side of the Rift Valley near Lake Victoria showed W-type, whereas specimens ($n = 77$) from Jaribuni, Magaoni, and Taveta located on the eastern side of the Rift Valley showed M-type.

Only two localities in Tanzania were analyzed. The first, Geita, is located on the western side of the Rift Valley near Lake Victoria, whereas Bagamoyo is on the coast, north of Dar-es-Salaam. RFLP typing of Bagamoyo ($n = 18$) revealed the M-type, whereas the specimens from Geita ($n = 48$) revealed MW-type found in southern Africa.

Malawi is located at the southern tip of the Rift Valley and specimens from two localities were analyzed. Karonga is located in northern Malawi, and all 36 specimens analyzed revealed the Y-type. Nkhotakota is located in the center of the country next to Lake Malawi (Fig. 3), and this population revealed the largest variation found to date, despite the small sample size ($n = 9$). Z-type was found in one specimen, six specimens had the M-type, one specimen had the W-type, and one specimen showed the Y-type.

The results presented here as well as those published in Garros et al. (2004a) were mapped on Fig. 3 to have a global picture of the RFLP genotypic distribution within *An. funestus* populations over the African continent.

Discussion

The Rift Valley is a phenomenal geographical structure in the eastern part of Africa extending north to south from Ethiopia to Malawi. Population geneticists have studied the effects of the Rift Valley on mosquito population structure of *An. gambiae* and *An. funestus* in Kenya and have recorded a lack of gene flow in localities separated by the Rift Valley (Lehmann et al. 2000, 2003; Braginets et al. 2003; Michel et al. 2005a).

The results presented here indicate that the Rift Valley plays a role in determining the RFLP profiles in various *An. funestus* populations in Africa. Specimens collected from the west side of the Rift Valley showed the W-type profile, and those on the east side of the Rift the M-type. However, Geita (Tanzania) on the west side of the Rift showed the MW-profile found in southern Africa and not W-type as found further north in Kenya. This difference might be because Geita is located on the southern side of Lake Victoria, which isolates it from other localities. To better understand the distribution of the different genotypes, it will be necessary to analyze more specimens from Tanzania around Lake Victoria by using various markers such as microsatellites, cytogenetics, and mtDNA.

The presence of two additional RFLP-types needs to be investigated further to determine the importance/relevance of these profiles. The only possible explanation for the difference in band intensities observed in Y-type, might be that the ratio of M- and W-types are not evenly distributed in the genome, because the rDNA is found in multiple copies within the genome (Cornel and Collins 1996). Cloning and sequencing the D3 fragment will answer this question. This approach was used by Garros et al. (2004) to study the MW-type from Mozambique and revealed that these clones had both copies of M and W within their genome; however, the W-type was dominant. However, other possible explanations, such as different PCR efficiencies because of differences in the DNA sequence need to be investigated to understand this phenomenon better.

Z-type is unique and was found in only four of 606 specimens. Blast analysis (http://www.ncbi.nlm.nih.gov/BLAST/Blast.cgi) revealed high similarity to *An. gambiae* cDNA sequence (accession number BX029704) as well as the 28S sequence of *An. rivulorum* and *An. minimus* C. Initial concern was that this was potential contamination from *An. gambiae.* However when the DNA was used in the species-specific PCR assay for *An. gambiae* (Scott et al. 1993), no amplicons were obtained, even after several attempts. The same DNA showed *An. funestus* diagnostic bands when using the species identification method (Koekemoer et al. 2002). Negative controls showed no amplicons in both the PCR MasterMix as well as the extraction negative control. Sequence alignment between MW-, Y-, and Z-types revealed the major difference between Z-type and the other two types (Fig. 2A). Figure 2B shows the close phylogeny between the MW- and Y-type.

Michel et al. (2005b) showed that microsatellite allele frequencies divided the *An. funestus* populations into three subdivisions: east, west, and central. The eastern population included populations from coastal Tanzania, Mozambique, and Malawi. The western division included populations from Burkina Faso, Mali, Nigeria, and western Kenya and the central division included populations from Gabon and coastal Angola. The RFLP types recorded here supports the west subdivision where we analyzed samples from Ghana, western Kenya, Mali, Nigeria, and Cote d'Ivoire. All these populations gave the W-type except one specimen from western Kenya that showed the Z-type. Samples from Malawi, coastal Tanzania, and Mozambique (east subdivision) did not cluster together, and RFLPs of all four types were present. Michel et al. (2005b) reported that samples from southwestern Uganda showed overlap with the three subdivisions. RFLP analysis from Uganda revealed no difference and all were typed as W-type.

Analysis using mtDNA revealed similar results as found by microsatellite analysis; however, in addition it revealed the presence of cryptic subdivisions within *An. funestus* (Michel et al. 2005b). One clade was present in all populations; however, *An. funestus* from Mozambique and Madagascar revealed the presence of an additional unique clade not present in the other populations. The RFLP analysis showed no unique types among these populations. This might be because of the RFLP analysis (genomic DNA) is not sensitive enough to reveal the differences observed in mtDNA.

The results presented here support the presence of population structuring within *An. funestus* and support previously published population genetic studies showing differentiation of populations from the eastern and western side of the Rift Valley. An in-depth population study is needed to establish the range of these populations and to determine assortative mating in sympatry. However, the significance of the RFLP types in *An. funestus* populations will have to be investigated and should be done in combination with detail morphology, cytogenetic and molecular analyses, ecology, and behavior.

Three clusters were identified: one cluster for West and central Africa (RFLP type-W), one cluster for East Africa and Madagascar (RFLP type-M), and one cluster for Angola, Mozambique, and South Africa (RFLP type-MW). In addition, Y-type was recorded in Malawi, whereas the fifth type, Z-type, was recorded from four localities (Angola, Malawi, Ghana, and Zambia).

Acknowledgments

We thank M. M. Weeto and T. Nkomo for technical assistance. We thank the following for supplying specimens used in this study: B. Ntomwa and the field teams of the Namibian Malaria Control Programme; Anglogold/Ashanti health services in Obuasi, Ghana; J. A. Chiphwanya for Malawi collections; K. Hargreaves, J. P. la Grange, and field teams from the malaria control programs in South Africa, S. Awolola, Nigeria; S. U. Nkunika, National Malaria Control Programme, Zambia; and T. Byembabazi and G. Egitat, Malaria Control Program and Vector Control Division, Kampala, Uganda. Partial funding was obtained from a PAL+2002 grant from the French Ministry of Research to S.M; UNDP/World Bank/WHO Special Programme for Research and Training in Tropical diseases (TDR,A20330LLK, A30345LK), National Research Foundation and South Africa Medical Research Council to L.L.K.

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Received 22 March 2006; accepted 3 August 2006.